1. Comparing two groups

Two-sample problems are among the most common in statistics.

Non-parametric tests Nonparametric statistical methods are inferential methods that do not assume a particular form distribution, such as the normal distribution, for the population distribution. Nonparametric methods are especially useful in two cases:

- When the data are ranks for the subjects, rather than quantitative measures.
- When it's inappropriate to assume normality, and when the ordinary statistical method is not robust to violations of the normality assumption. For instance, we might prefer not to assume normality because we think the distribution will be skewed, non-normal or contain outliers. Or, perhaps we have no idea about the distribution shape, and the sample size is too small to give us much information about it.

One kind of non-parametric tests are **rank tests**, based on the rank (place in order) of each observation in the set of all the data. The kind of assumptions required, the nature of the hypotheses tested, the big idea of using ranks, and the contrast between exact distributions for use with small samples and approximations for use with larger samples are common to all rank tests. In this document, four rank tests are mentioned: the **Wilcoxon rank sum test**, the **sign test**, the **Wilcoxon Signed rank test** and the **Kruskal-Wallis test**.

1.1. Independent samples

Most comparisons of groups use independent samples from the groups. The observations in one sample are independent of those in the other sample. For instance, randomized experiments that randomly allocate subjects to two treatments have independent samples. Another example is visitors to a web site that are routed to a version A or B with the same functionalities. Another way independent samples occur is when an observational study separates subjects into groups according to their value for an explanatory variable, such as smoking status or gender. If the overall sample was randomly selected then the groups can be treated as independent random samples.

1.1.1. Quantitative response variable

a) (Parametric) The two-sample unpaired t-test is a commonly used test that compares the means of two samples. The appropriate data to use this test exhibits the following characteristics:

   - Two-sample data. That is, one measurement variable in two groups or samples.
   - The dependent variable is interval/ratio, and is continuous.
   - The independent variable is a factor with two levels. That is, two groups.
   - Data for each population are normally distributed.
   - For Student's t-test, the two samples need to have the same variance. However, Welch's t-test, which is used by default in R, does not assume equal variances.

- Observations between groups are independent. That is, not paired or repeated measures data.
- Moderate skewness is permissible if the data distribution is unimodal without outliers

The hypotheses being tested are:

- Null hypothesis: The means of the populations from which the data were sampled for each group are equal.
- Alternative hypothesis (two-sided): The means of the populations from which the data were sampled for each group are not equal.

Reporting significant results as "Mean of variable Y for group A was different than that for group B" is acceptable.

The function `t.test()` in R with argument `paired=FALSE` performs the test. Set `var.equal =FALSE` to perform Welch's test.

Example: to compare the mean driving reaction time of drivers in two driving situations, a control group and a group carrying out a phone conversation. 64 students were randomly assigned to one of the groups. This is an experiment but you could have data from an observational study as well.

b) (Non-parametric, rank test) When classical statistical assumptions are not met you can use non-parametric methods, for instance, the Wilcoxon Rank Sum Test which is a rank-based test that compares values for two groups. This test converts observations to ranks when the response is not itself a rank. A significant result suggests that the values for the two groups are different. Without further assumptions about the distribution of the data, this test does not address hypotheses about the medians of the groups. Instead, the test addresses if it is likely that an observation in one group is greater than an observation in the other. This is sometimes stated as testing if one sample has stochastic dominance compared with the other.

This test is equivalent to the test called *two-sample Mann-Whitney U test.*

The appropriate data to use this test on exhibit the following characteristics:

- Two-sample data. That is, one-way data with two groups only.
- The dependent variable is ordinal, interval, or ratio.
- The independent variable is a factor with two levels. That is, two groups.
- Observations between groups are independent. That is, not paired or repeated measures data.
- In order to be a test of medians, the distributions of values for each group need to be of similar shape and spread. Otherwise the test is typically a test of stochastic equality.

The hypotheses tested are:

- Null hypothesis: The two groups are sampled from populations with identical distributions. Typically, that the sampled populations exhibit stochastic equality.
- Alternative hypothesis (two-sided): The two groups are sampled from populations with different distributions. Typically, that one sampled population exhibits stochastic dominance, the values in one sample are systematically higher or lower than the values in the other sample.

Significant results can be reported as e.g. "Values for group A were significantly different from those for group B." for a two-sided alternative.

Large sample p-values are easily obtained using a Normal distribution. The null hypothesis is that of identical population distributions for the two groups. This implies equal population sum ranks (or mean ranks). You can use R function `wilcox.test()` in R with argument `paired=FALSE` or function `wilcox_test()` from R package `coin`.

c) (Non-parametric, simulation based) Comparing two groups with a permutation test (more about this procedure at the end of this file).

1.1.2. Categorical response variable

χ^2 test of homogeneity (or independence, depending on the setting although both are equivalent). It is equivalent to test if the proportions are equal in each category (of the explanatory variable). For instance, comparing a sample of smokers to a sample of non smokers (explanatory variable) in terms of their severity of asthma. The data consists of several independent samples, one for each group. The data can be presented arranged in a contingency table. For example, counts of fatalities in automobile accidents in a recent year using as an explanatory variable whether people wore the seatbelt. The question is: Are the two variables independent? Is wearing the seatbelt related to a higher chance of survivability in a car accident?

Wore seat belt -Outcome	Survived	Died	Total
Yes	412,368	510	412878
No	162,257	1601	164,128

The test works computing the expected counts in the table under the hypothesis of independence and comparing them to the observed counts. For the χ^2 approximation to work, it is necessary for the observed counts in the cells to be ≥ 3 and for the expected counts to be ≥ 5. When these assumptions are not met you can specify in R's function `chisq.test` to compute the p-value by Monte Carlo simulation (option `simulate.p.value=TRUE`).

1.2. Dependent samples

Sometimes we want to compare treatments or conditions at the individual level. These situations produce two samples that are not independent — they are related to each other. For instance, when the two samples involve the same subjects, they are dependent. In general, anytime we measure any subject twice (before and after a treatment, for example) leads to dependent samples. Dependent samples also result when the data are matched pairs, each subject in one sample is matched with one subject in the other sample. An example is a set of married couples, the men being in one sample and the woman in the other. Another example is when it is used people matched for age, sex and education in social studies to allow canceling out the effect of these potential lurking variables. Data from dependent samples need different statistical methods than data from independent samples.

1.2.1. Quantitative response variable

a) (Parametric) It compares the means of two populations of paired observations by testing if the difference between pairs is statistically different from zero.

   The appropriate data to use this test exhibits the following characteristics:

   - Two-sample data. That is, one measurement variable in two groups or samples
   - Dependent variable is interval/ratio, and is continuous
   - Independent variable is a factor with two levels. That is, two groups
   - Data are paired. That is, the measurement for each observation in one group can be paired logically or by subject to a measurement in the other group
   - The distribution of the difference of paired measurements is normally distributed
   - Moderate skewness is permissible if the data distribution is unimodal without outliers

   The hypotheses tested are:

   - Null hypothesis: The population mean of the differences between paired observations is equal to zero.
   - Alternative hypothesis (two-sided): The population mean of the differences between paired observations is not equal to zero.

   Reporting significant results as "Mean of variable Y for group A was different than that for group B." or "Variable Y increased from before to after" is acceptable.

   To perform this test in R, simply calculate difference scores for the paired observations and then use methods for a single sample to test if that mean difference is 0. Or use unction `t.test()` on the original data with argument `paired=TRUE`.

b) (Non-parametric, rank test) The sign test for two sample paired data (**sign test for matched pairs**). The assumption is that we have a random sample of matched pairs for which we can evaluate which observation in a pair has the better response. For such a matched pairs experiment, let p denote the population proportion of cases for which a particular treatment does better than the other treatment. Under the null hypothesis of identical treatment effects, $p = 0.50$. That is, each treatment should have the better response outcome about half the time. A test that compares matched pairs in this way is called a sign test. The name refers to how the method evaluates for each matched pair whether the difference between the first and second response is positive or negative. It uses the information about which response is higher and how many responses are higher, not the quantitative information about how much higher. For instance, a sample of students report the time they spend watching TV and browsing the Internet. We are interested in answering the question: Which do most students spend more time doing—browsing the Internet or watching TV?

The appropriate data to use this test exhibits the following characteristics:

- Two-sample paired data. That is, one-way data with two groups only, where the observations are paired between groups.
- Dependent variable is ordinal, interval, or ratio.
- Independent variable is a factor with two levels. That is, two groups.

The hypotheses tested are:

- Null hypothesis: H_0 : population proportion who have better response for a particular group $p = 0.5$.
- $H_1 : p \neq 0.5$ (two sided) or $H_1 : p > 0.5$ or < 0.5 (one sided).

For large n, being n the sample number of pairs of observations for which the two responses differ, the p-value can be computed using the normal distribution. For small n the sign test can be conducted using the binomial distribution.

c) (Non-parametric, rank test) **The Wilcoxon signed-ranks test**. With matched pairs data, for each pair the sign test merely observes which treatment does better, but not how much better. The Wilcoxon signed-ranks test is a nonparametric test designed for cases in which the comparisons of the paired observations can themselves be ranked. For each matched pair of responses, it measures the difference between the responses. It tests the hypothesis, H_0 : population median of difference scores is 0. The test uses the quantitative information provided by the n difference scores by ranking their magnitudes, in absolute value.
Although the Wilcoxon signed-ranks test has the advantage compared to the sign test that it can take into account the *sizes* of the differences and not merely their *sign*, it also has a disadvantage. It makes an additional assumption: the population distribution of the difference scores must be symmetric. This is hard to prove when the sample size is very small.

1.2.2. Categorical response variable

When the response variable is categorical, McNemar's test for comparing proportions with matched-pair data is used. Let's label the four cells of a 2×2 contingency table as a, b, c and d.

a	b
c	d

For instance, imagine we ask the same people twice whether they believe in Heaven and in Hell. We get the following table:

	Belief in Hell	
Belief in Heaven	Yes	No
Yes	a	b
No	c	d

For testing $H_0 : p_1 = p_2$, that is an equal proportion believes in Heaven and Hell, we can use the test statistic:

$$z = \frac{b - c}{\sqrt{b + c}}$$

It has an approximate standard normal distribution when H_0 is true. To use this test, the sum of the two off-diagonal counts b and c should be at least 30, but in practice the two-sided test works well even if this is not true. The previous test statistic z is equivalent to use z^2, which follows a χ^2_1

For example, filling the previous table with a sample of adults in the U.S. in which 1314 subjects were asked two questions: Do you believe in Heaven? and Do you believe in Hell? we get:

	Belief in Hell		
Belief in Heaven	Yes	No	Total
Yes	955	162	1117
No	9	188	197
Total	964	350	1314

We estimate p_1, the proportion who believe in Heaven as $\hat{p_1} = \frac{1117}{1314}$ and p_2 the proportion who believed in Hell as $\hat{p_2} = \frac{964}{1314}$. The result of McNemar's test gives an extremely low p-value. Thus, there is very strong evidence that the proportion of U.S. adults believing in heaven differs from (in fact is larger than) the proportion believing in hell.

2. Comparing two groups with a permutation test

A permutation test (a simulation based test) starts from a null hypothesis of equal population distributions. This null hypothesis goes beyond just requiring the population means to be the same and is a stronger statement of "no effect."

$$H_0 : \text{ population distribution in Group 1} = \text{population distribution in Group 2}$$

This test derives the sampling distribution by considering all possible (or a large sample of random) permutations of the original data. A permutation is one particular way of dividing the (n_1+n_2) observations into two groups of size n_1 and n_2, respectively. For each such permutation, the test statistic (e.g., the difference in the sample means) is computed from the resulting dataset. The key ingredients of a permutation test to compare two groups are the following:

- Assumptions
  - A quantitative response variable observed in each of two groups
  - Independent random samples, either from random sampling or a randomized experiment. For an experiment, the assumption of a random sample of subjects is not necessary as long as the subjects are assigned to the two groups randomly.
- Hypotheses

- H_0 : *identical population distributions for the two groups (this implies equal population means).*
- H_1 : *Population means differ between the two groups. One-sided versions are also possible.*
- H_1 could also be in terms of other population parameters, such as population medians.

▪ Test statistic: the difference $(\bar{x_1} - \bar{x_2})$ of the sample means. If the alternative hypothesis is in terms of medians, then the test statistic is the difference in the sample medians.

▪ P-value: The P-value is the proportion out of all possible permutations with a test statistic as extreme or even more extreme than the observed test statistic. When the number of all possible permutations is too large, a random sample of a large number of permutations (such as $10,000$) is considered. Then, the P-value is the proportion out of these $10,000$ random permutations that give a test statistic as extreme or even more extreme than the observed one.

Example: Symptom free days of allergy patients on High and Low doses of an experimental drug. The data looks like this:

Dose	Days w/o symptoms
High	12
High	13
⋮	⋮
Low	6
Low	7

I'll explain the procedure in class.

3. McNemar test: an example

This is an example of matched-pairs with a categorical response which leads to what is known as a square table. Suppose that we ask a group of $n = 1600$ voters whether they approve of the President's performance at two points in time. The two characteristics are approval of the President's performance (Presidential approval: approve & disapprove) at two periods of time (Time 1 and Time 2). It is only sensible to assume that these two variables should be related, because the same 1600 voters are being asked the same question at two points in time. An individual's political views at one time are likely to be somewhat similar to his or her views at another time. Has the president's approval rating changed over time? This is the same as asking if the margins of the table are the same, which can be done with the test of marginal homogeneity or with McNemar's test. Here, the latter is illustrated.

Time 1 -Time 2	Approved	Disapproved	Total
Approved	794	150	944
Disapproved	86	570	656
Total	880	720	1600

Entering the data in R and output of the test:

```
datos <- matrix(c(794, 86, 150, 570),
                nrow = 2,
                dimnames = list("Approved" = c("Yes", "No"),
                                "Disapproved" = c("Yes", "No")))
mcnemar.test(datos, correct=FALSE)
McNemar's Chi-squared test

data:  datos
McNemar's chi-squared = 16.818, df = 1, p-value = 4.115e-05
```

There is very strong evidence that the probability of approval at the time of the first survey was different (higher than) that the time of the second survey.

4. McNemar's test: another example

Speech recognition system: Research in comparing the quality of different speech recognition systems uses a series of isolated words as a benchmark test, finding for each system the proportion of words for which an error of recognition occurs. The table below shows data from one of the first articles that showed how to conduct such a test. The article compared speech recognition systems called generalized minimal distortion segmentation (GMDS) and continuous density hidden Markov model (CDHMM). Conduct McNemar's test of the null hypothesis that the probability of a correct outcome is the same for each system.

Method 1 -Method 2	Correct	Incorrect	Total
Correct	1921	58	1979
Incorrect	16	5	21
Total	1937	63	2000

The two-sided P-value is 0.00004. There is extremely strong evidence against the null hypothesis that the correct detection rates are the same for the two systems.

5. With more than two groups

5.1. Independent samples

5.1.1. One-way ANOVA

The analysis of variance method compares means of several groups. Let g denote the number of groups. Each group has a corresponding population of subjects. The means of the response variable for the g populations are denoted by $\mu_1, \mu_2, \ldots, \mu_g$. The analysis of variance is a significance test of the null hypothesis of equal population means:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_g$$

The alternative hypothesis is:

$$H_1 : \text{at least two of the population means are unequal}$$

If H_0 is false, perhaps all the population means differ, but perhaps merely one mean differs from the others. The test analyzes whether the differences observed among the sample means could have reasonably occurred by chance, if the null hypothesis of equal population means were true. The assumptions for the ANOVA test comparing population means are as follows:

- The population distributions of the response variable for the g groups are normal, with the same standard deviation for each group.
- Randomization (depends on data collection method): In a survey sample, independent random samples are selected from each of the g populations. For an experiment, subjects are randomly assigned separately to the g groups.

The ANOVA test is a test of independence between the quantitative response variable and the group factor. The test uses the F distribution and its steps are the following:

- Assumptions: Independent random samples (either from random sampling or a randomized experiment), normal population distributions with equal standard deviations
- Hypotheses: $H_0 : \mu_1 = \mu_2 = ... = \mu_g$ (Equal population means for g groups), H_1 : at least two of the population means are unequal.
- Test statistic:
$$F = \frac{\text{Between-groups variability}}{\text{Within-groups variability}}$$
  F sampling distribution has $df_1 = g - 1, df_2 = N - g$.
- P -value: Right-tail probability of above observed F value.
- Conclusion: Interpret in context. If decision needed, reject H_0 if p-value $\leq$ significance level (such as 0.05).

Since the ANOVA F test is robust to moderate breakdowns in the population normality and equal standard deviation assumptions, in practice it is used unless (1) graphical methods show extreme skew for the response variable or (2) the largest group standard deviation is more than about double the smallest group standard deviation and the sample sizes are unequal.

5.1.2. Kruskal-Wallis test

Just like comparisons of means for two groups extend to comparisons for many groups, using the analysis of variance (ANOVA) F test, the Wilcoxon test for comparing mean ranks of two groups extends to a comparison of mean ranks for several groups. This rank test is called the Kruskal-Wallis test. As in the Wilcoxon test, the Kruskal-Wallis test assumes that the samples are independent random samples, and the null hypothesis states that the groups have identical population distributions for the response variable.

When would we use this test? The ANOVA F test assumes normal population distributions. The Kruskal-Wallis test does not have this assumption. It's a safer method to use with small samples when not much information is available about the shape of the distributions. It's also useful when the data are merely ranks and we don't have a quantitative measurement of the response variable.

5.2. Dependent samples

5.2.1. One way ANOVA with repeated measures

An ANOVA with repeated measures is used to compare three or more group means where the participants are the same in each group. This usually occurs in two situations: (1) when participants are measured multiple times to see changes to an intervention; or (2) when participants are subjected to more than one condition/trial and the response to each of these conditions wants to be compared. The name repeated measures stems from the fact that all treatments are administered to each unit. For instance, you might be investigating the effect of a 6-month exercise training programme on blood pressure and want to measure blood pressure at 3 separate time points (pre-, midway and post-exercise intervention) for the same group of participants.

5.2.2. The Friedman rank sum test

A non parametric version of the previous one.

All the tests mentioned in this section, with more than two groups, are of use with a quantitative response variable. When the response is categorical, you can use the χ^2-test for independent samples. With dependent samples, a categorical response variable and more than two groups, you can use Cochran's Q-test.